But is that really all?

Michael: fields and light

Newton knew that his equations did not describe *all* the forces that exist in nature. There are forces other than gravity that act upon bodies. Things don't move just when they fall. The first problem left open by Newton was to understand the *other* forces that determine what happens around us. This had to wait until the nineteenth century, and it led to two surprises.

The first surprise is that almost all phenomena we see are governed by *a single* force, other than gravity: the force that today we call electromagnetism. It is this force which holds together the matter that forms solid bodies; holds together atoms in molecules, and electrons in atoms. This is what makes chemistry and living matter work. It is this force which operates in the neurons of our brain and governs our processing of the information on the world we perceive, and the way we think. And it's always this force that creates the friction which stops a sliding object, which softens the landing of a parachutist, which turns electric motors and combustion engines,[fn7] or that allows us to turn on lights and listen to the radio.

The second and biggest surprise, crucial to the story I'm telling, is that understanding this force requires an important modification to the world of Newton: the modification out of which modern physics was born, and the most important notion to keep in focus, to understand the rest of this book – the notion of field.

The understanding of how electromagnetic force works was made by another Briton, or rather by two: science's oddest couple – Michael Faraday and James Clerk Maxwell.

Michael Faraday is an impoverished Londoner without formal education, who works first in a bookbindery, then in a laboratory, where he excels, gains his master's confidence and grows into the most brilliant experimenter of nineteenth-century physics and its greatest visionary. Without knowing mathematics, he writes one of the best books of physics ever written, virtually devoid of equations. He sees physics with his mind's eye, and with his mind's eye creates worlds. James Clerk Maxwell is a rich Scottish aristocrat and one of the greatest mathematicians of the century. Despite being separated by a gulf in intellectual style as well as social origin, they succeed in understanding each other – and, together, combining two kinds of genius, they open the way to modern physics.